33 An Introduction to Invertebrates

Figure 33.1 The blue dragon (*Glaucus atlanticus*) has thin, finger-like structures that help it to float (upside down) on the sea's surface. It eats Portuguese men-of-war and absorbs their stinging cells, which it then uses to deliver a deadly sting of its own. The blue dragon is just one of more than a million species of invertebrates (animals without a backbone), which account for over 95% of animal species.

KEY CONCEPTS

Study Tip

Make a table: Invertebrates differ in how they obtain food, remove wastes, reproduce, and move throughout their habitat. As you read, summarize how members of the 11 phyla of invertebrates described in detail in this chapter solve these shared challenges of life.

Phylum	Feeding method	Waste removal	Repro-duction	Movement
Porifera	Filter feeders	Diffusion	Herma-phrodites	Sessile

Go to Mastering Biology

For Students (in eText and Study Area)
- Get Ready for Chapter 33
- Video: Echinoderm Tube Feet
- HHMI Animation: Coral Bleaching

For Instructors to Assign (in Item Library)
- Tutorial: Ecdysozoans
- Scientific Skills Exercise: Understanding Experimental Design and Interpreting Data

How can we make sense of the great number and morphological diversity of invertebrates?

Classifying invertebrate species into groups based on **evolutionary relationships**—as in this simplified phylogeny of all animals—helps us to understand the great diversity of invertebrate life.